Ajara Oyinloye Lawal

Data Scientist | MS Engineering Artificial Intelligence, Carnegie Mellon University Africa

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Profile

I'm a data scientist finishing an MS in Engineering Artificial Intelligence at Carnegie Mellon University Africa (December 2026). My work spans computer vision, multi-agent LLM systems and data pipelines, and I am drawn to problems where AI has to work for people with few resources: search-and-rescue drones, flood and drought early warning, and checking whether language models give advice people can actually follow. Before graduate school I built internal systems at Ghana's national ICT centre and ran operations for a digital agency, and teaching young people to code has run alongside all of it.

Education

MS in Engineering Artificial Intelligence — Aug 2025 – Dec 2026 (expected)

Carnegie Mellon University Africa, Kigali, Rwanda

- **Coursework:** Machine Learning, Introduction to Deep Learning, Applied Computer Vision, Data Science, AI Systems Design

BSc Computer Science (Honours) — Sep 2019 – Sep 2023

University for Development Studies, Navrongo, Ghana | Mastercard Foundation Scholar (CAMFED)

Experience

Summer Intern, RISE Project (Resilience in Secondary Education) — Jun 2026 – Aug 2026

Carnegie Mellon University Africa, Kigali

- Wrote, organised and updated interactive course content on OLI Torus, integrated with Rwanda's Moodle-based REB learning platform, for students working toward the Advanced General Certificate of Secondary Education.

Student Affairs Support — Sep 2025 – May 2026

Carnegie Mellon University Africa, Kigali

- Handled student communications, calendars and day-to-day administration in Google Workspace for more than 100 graduate students.
- Coordinated logistics for campus events, from sourcing vendors and placing orders to keeping budget and expenditure records in Excel and Google Sheets.

Junior Software Developer (National Service) — Nov 2023 – Dec 2024

Ghana-India Kofi Annan Centre of Excellence in ICT (GI-KACE), Accra, Ghana

- Co-developed SmartConference in Python FastAPI and PostgreSQL, replacing Google Forms registration and paper sign-in sheets with one app where attendees find their name, mark attendance and collect their name tag.
- Co-developed a Staff Appraisal ERP that moved appraisals off paper and onto a digital platform, working in Scrum sprints.
- Tested and validated teammates' backend APIs through Swagger.
- Co-facilitated classes in Python, Office Productivity and Data Analytics for students and working professionals.

Operations Manager (Intern, remote) — Nov 2021 – Mar 2025

Namibra, Kumasi, Ghana | digital agency building web, mobile, AI and BI solutions

- Introduced Jira and ClickUp to run the team's work in Scrum sprints; the team settled on ClickUp, and planning and tracking work became easier.
- Researched new tools for the business, chose the best fit for operations and trained the team to use them.
- Managed day-to-day activities so the organisation ran efficiently.

Industrial Attachment Intern — Oct 2021 – Dec 2021

CKT-University of Technology and Applied Sciences, Navrongo, Ghana

- Replaced network cables and switches, organised cabling, repaired internet ports and terminated Ethernet cables for the institution, and helped build its booking system.

Projects

Lightweight Human Detection for Search & Rescue Drones — Apr 2026

- Fine-tuned YOLOv8n/s and RetinaNet on aerial search-and-rescue imagery with a two-layer augmentation pipeline: Ultralytics runtime transforms plus custom offline degradation that mimics consumer-drone artefacts.
- Raised YOLOv8n to 94.31% mAP@0.5 at 178.5 FPS, 7.5 points above the published baseline with a model 20x smaller.
- Paired YOLOv8n with Depth Anything V2 and a polynomial calibration model to estimate distance to each person, at 2.04 m mean absolute error over 1–24.5 m, to support field triage.

Stateful Multi-Agent IT Support System — Mar 2026

- With a teammate, rebuilt a single-agent RAG IT-support tool as a Planner, Executor and Verifier system with persistent SQLite memory and adaptive control that retries tools, re-plans or escalates when metrics fall below set thresholds.

- Built an evaluation framework scoring groundedness, tool-selection accuracy, task completion, plan adherence and hallucination rate.
- Traced failing tool calls to a mismatch between LLM-generated and expected parameter names; explicit schemas and a normalisation layer lifted response quality from 0.50 to 0.75 and plan adherence above 0.85.

Fine-Tuning RetinaNet for Custom Object Detection Mar 2026

- Collected and annotated a custom image dataset in Roboflow under varied lighting, angles and backgrounds, exported in COCO format.
- Fine-tuned a COCO-pretrained RetinaNet (ResNet-50 + FPN) with Focal Loss in PyTorch on 141 images, reaching 1.00 mAP@0.5 and 0.936 mAP@0.5:0.95, ahead of YOLOv8, Faster R-CNN and EfficientDet baselines on the same data.

Resource-Assumption Bias in a Small Language Model (in progress) Oct 2026

- Designing an evaluation of Qwen3.5-4B for advice that assumes things people lack, such as a car, a fridge, internet or a nearby clinic: 30 scenarios from rural Ghana and Rwanda, each posed under four context conditions, scored for infeasible recommendations, local emergency numbers and sensitivity to context.

Medical AI Robustness (in progress) Oct 2026

- With three classmates, testing how medical AI models hold up when adversarial attacks are added to the data.

Explainable AI Early-Warning System for Flood and Drought Prediction in Africa 2025

- Built the data cleaning and preprocessing pipeline in Python, NumPy and Pandas, contributed to the explainable AI model and co-authored the research paper.

Forecasting International Homicide Rates: ML vs Statistical Time-Series Methods 2025

- Compared machine learning (Scikit-learn, LSTM) and statistical time-series models, evaluated them with RMSE and other metrics, and wrote the paper.

Leadership & Community

- **Treasurer, CMU-Africa Data Science Club** (Dec 2025 – Dec 2026): budget for events, seek sponsorship and outside support, and co-organise events with the executive team.
- **Country Representative, CMU Africa** (Dec 2025 – Dec 2026): represent Ghanaian students, welcome new arrivals, organise country hangouts and help with multicultural and diversity days.
- **Procurement Committee Secretary, SRC, CKT-UTAS** (2022 – 2023): evaluated bids and selected suppliers on value for money.
- **Computer Science Society:** Welfare Committee Chairperson, CKT-UTAS (2022 – 2023); Secretary (2021 – 2022) and Audit Committee Member (2019 – 2020), UDS.
- **STEM education:** taught young learners coding, game development, animation and internet safety through community initiatives.
- **Women and girls (CAMFED alumna):** taught women soap and bead making and mentored girls in basic school.

Certifications & Awards

- AWS Certified Cloud Practitioner (2024)
- Data Analytics with Python and Power BI, GI-KACE (2023)
- Virtual Assistant certificate, ALX Africa
- Mastercard Foundation Scholar through CAMFED, secondary (2015 – 2018) and tertiary (2019 – 2023)

Skills

Programming: Python (advanced), SQL, JavaScript, PHP, HTML/CSS
Machine learning & AI: PyTorch, TensorFlow, Scikit-learn; object detection (YOLOv8, RetinaNet, Faster R-CNN), segmentation (U-Net, Mask R-CNN, SegFormer, SAM), monocular depth estimation; LSTM and time-series forecasting; RAG and multi-agent LLM systems; model evaluation
Data: Pandas, NumPy, Matplotlib, Power BI, Excel, Google Sheets, Roboflow, dataset collection and annotation
Engineering: FastAPI, PostgreSQL, SQLite, Swagger, Git, Linux, AWS
Ways of working: Scrum, Design Thinking, Jira, ClickUp, teaching and mentoring
Languages: Yoruba (native), English (fluent), French (conversational)